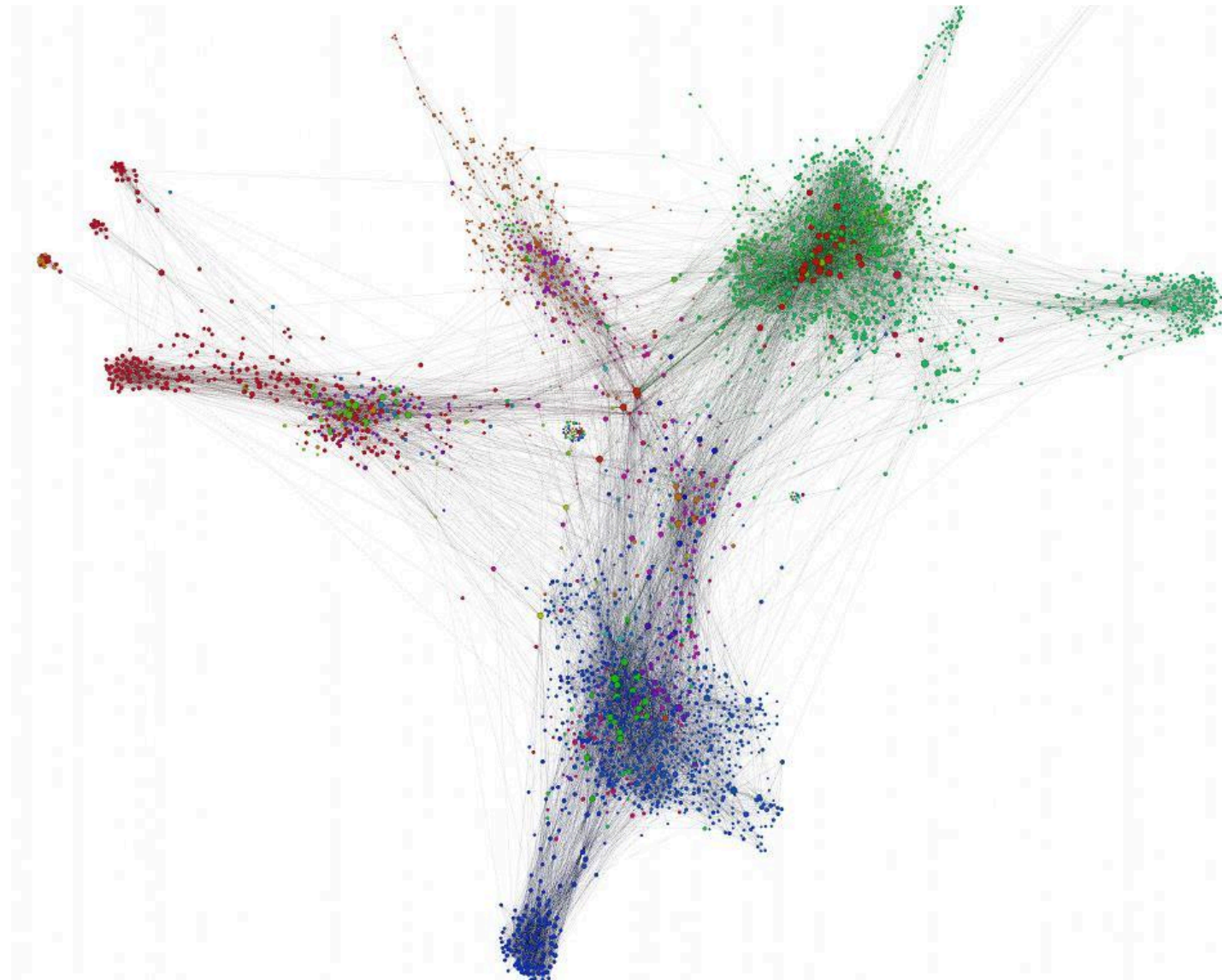
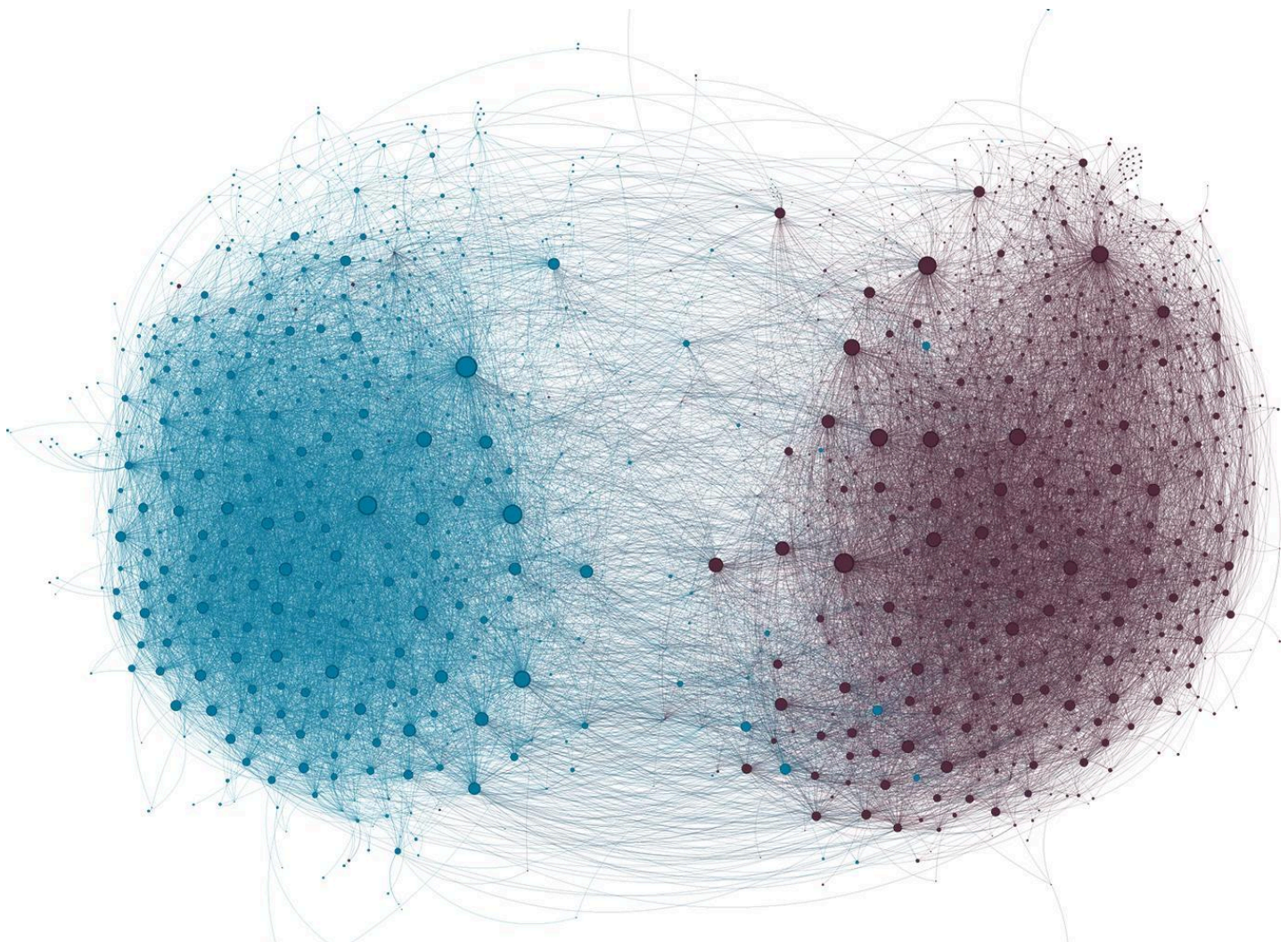
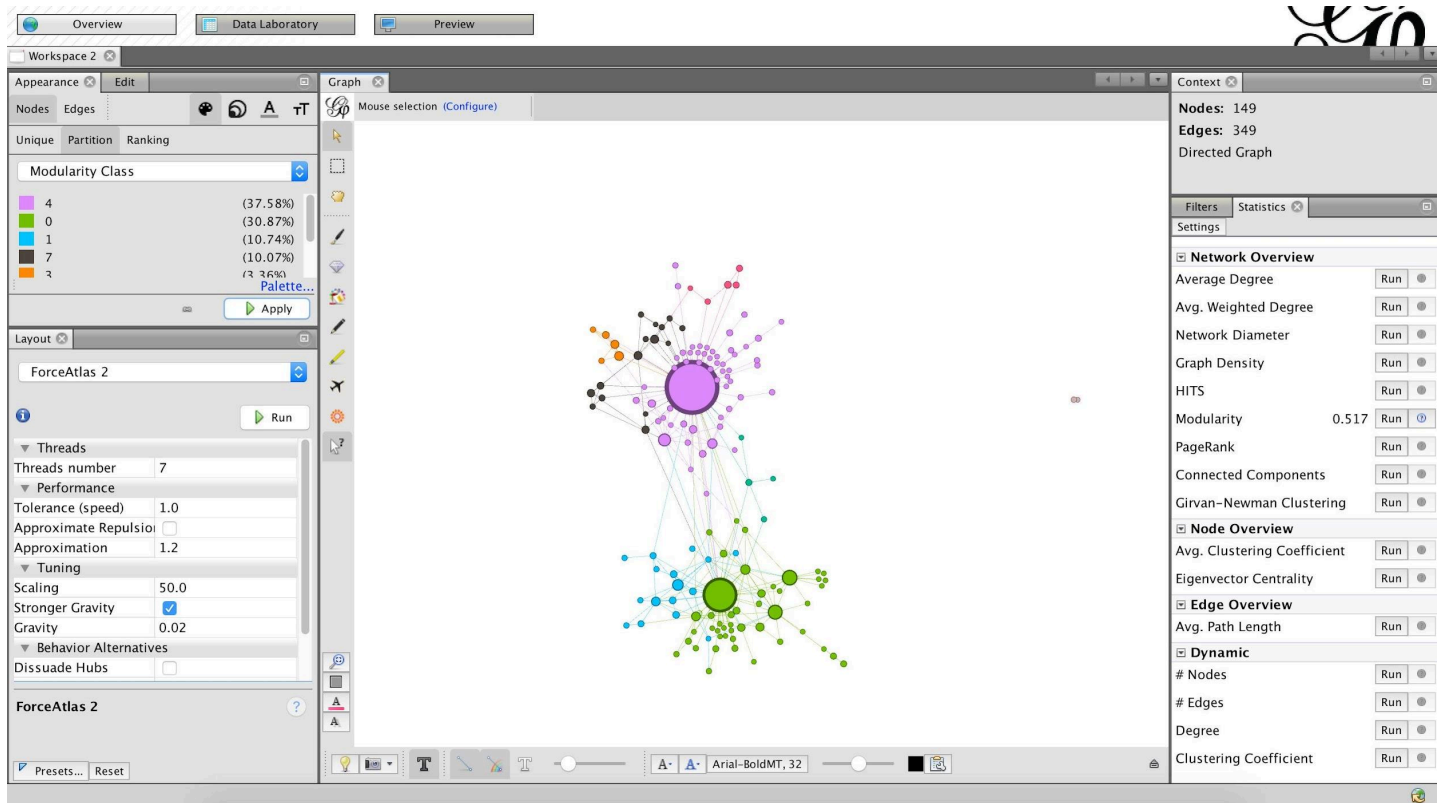


Gephi

Найпопулярніший desktop-tool для великих графів





Spline...

Apply

Layout x

ForceAtlas 2

Run

Threads

Threads number 7

Performance

Tolerance (speed) 1.0

Approximate Repulsion ☐

Approximation 1.2

Tuning

Scaling 100.0

Stronger Gravity ☐

Gravity 1.0

Behavior Alternatives

Dissuade Hubs ☐

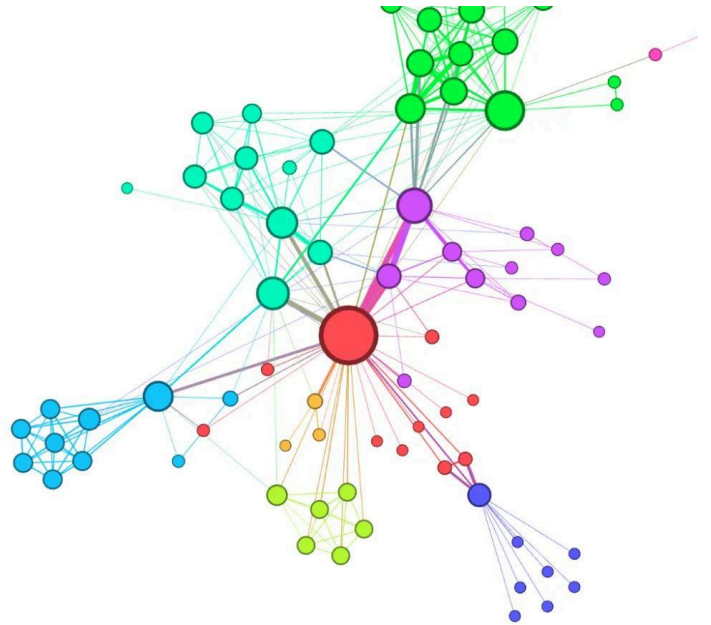
LinLog mode ☐

Prevent Overlap ☒

Edge Weight Influence 1.0

Normalize edge weights ☐

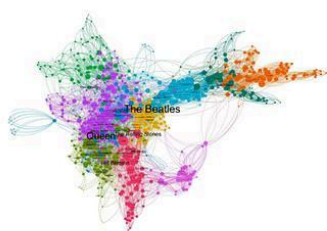
ForceAtlas 2



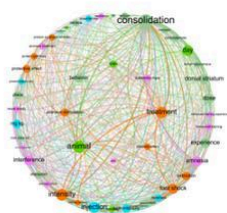
GEPHI NETWORK GALLERY



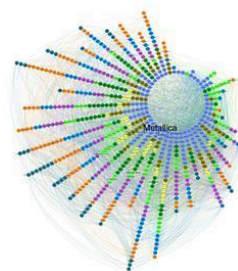
By Dr. Verónica Espinoza



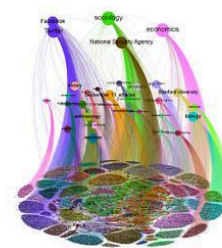
Force Atlas 2 Layout



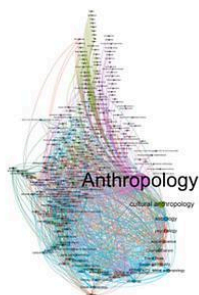
Dual Circle Layout



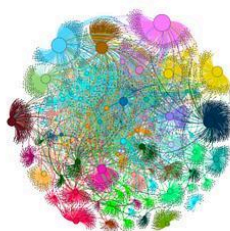
Radial Axis Layout



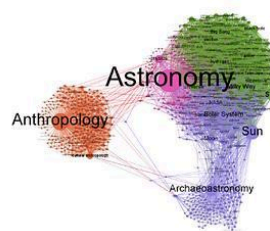
Fruchterman Reingold +
Network Splitter 3D



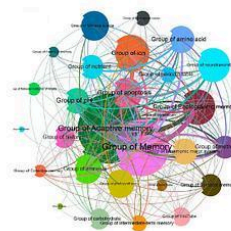
Radial Axis Layout



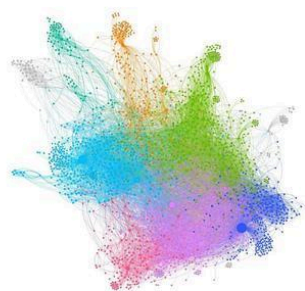
Force Atlas 2 Layout



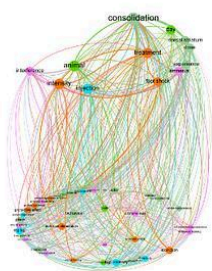
Force Atlas Layout + Append
to existing workspace



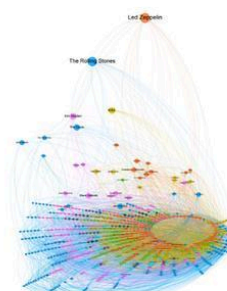
Fruchterman Reingold +
Groups by partition



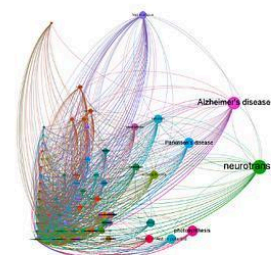
Force Atlas Layout



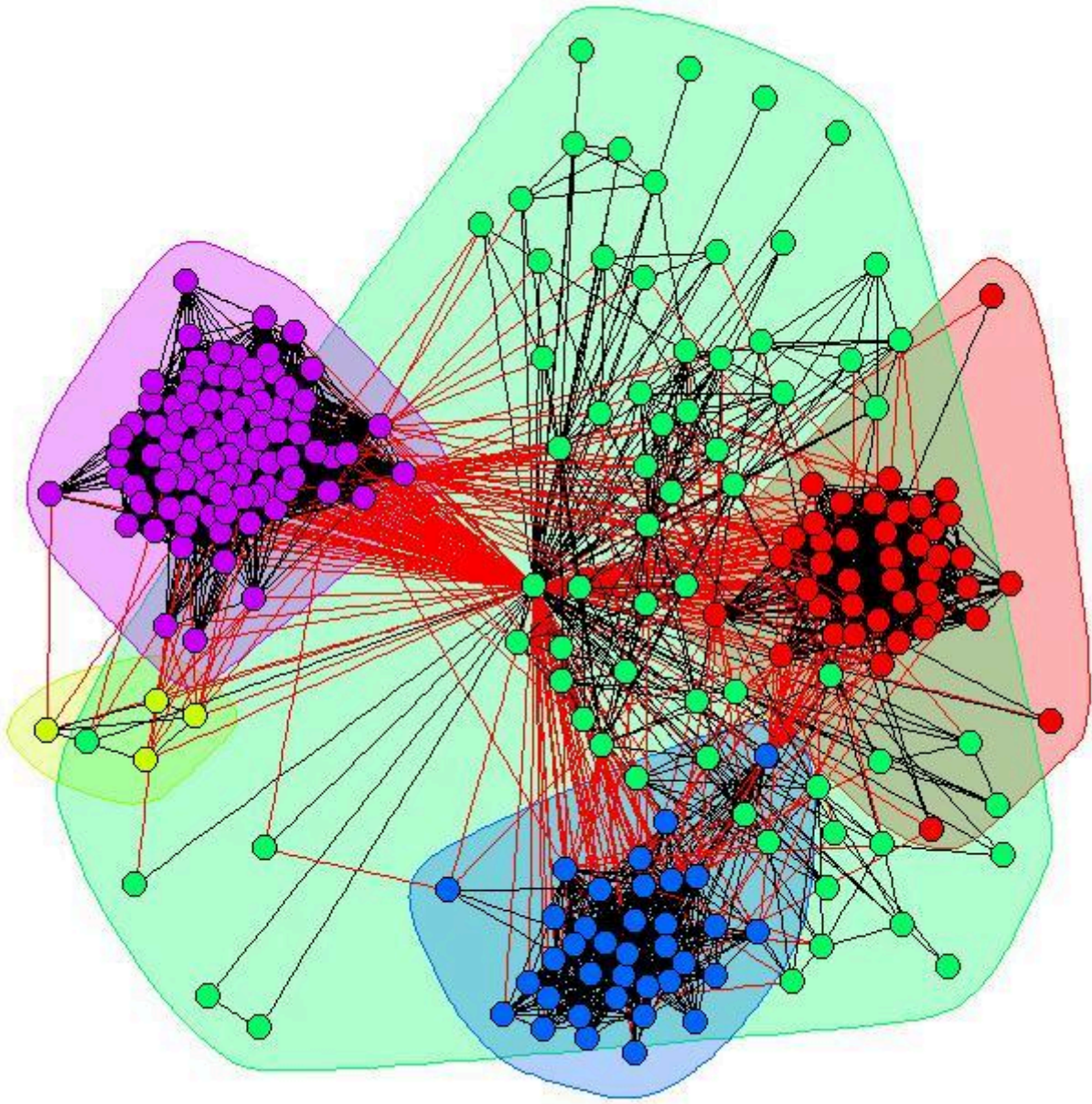
Dual Circle Layout +
Network Splitter 3D

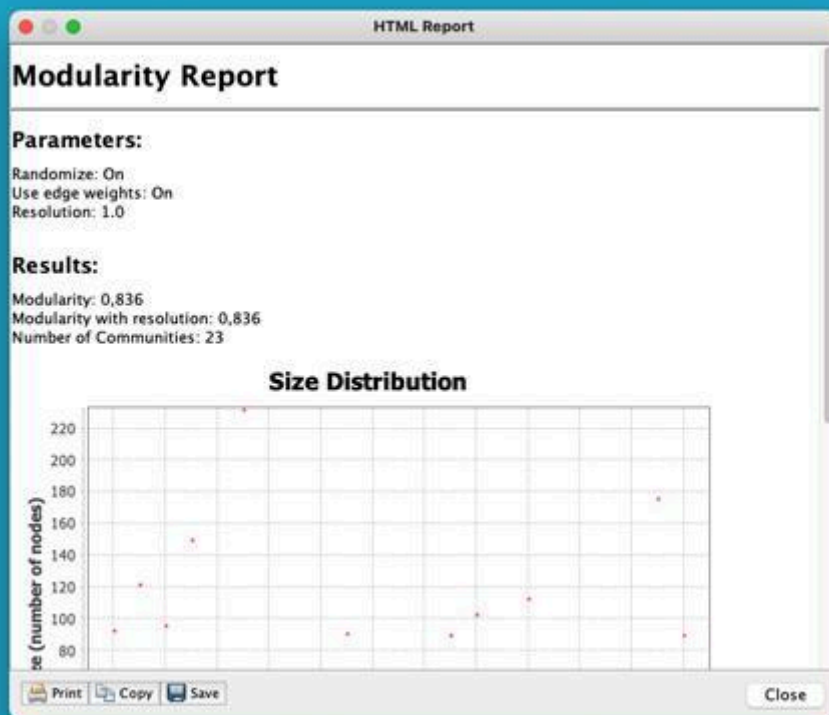


Radial Axis Layout +
Network Splitter 3D



Position Ranking Plugin

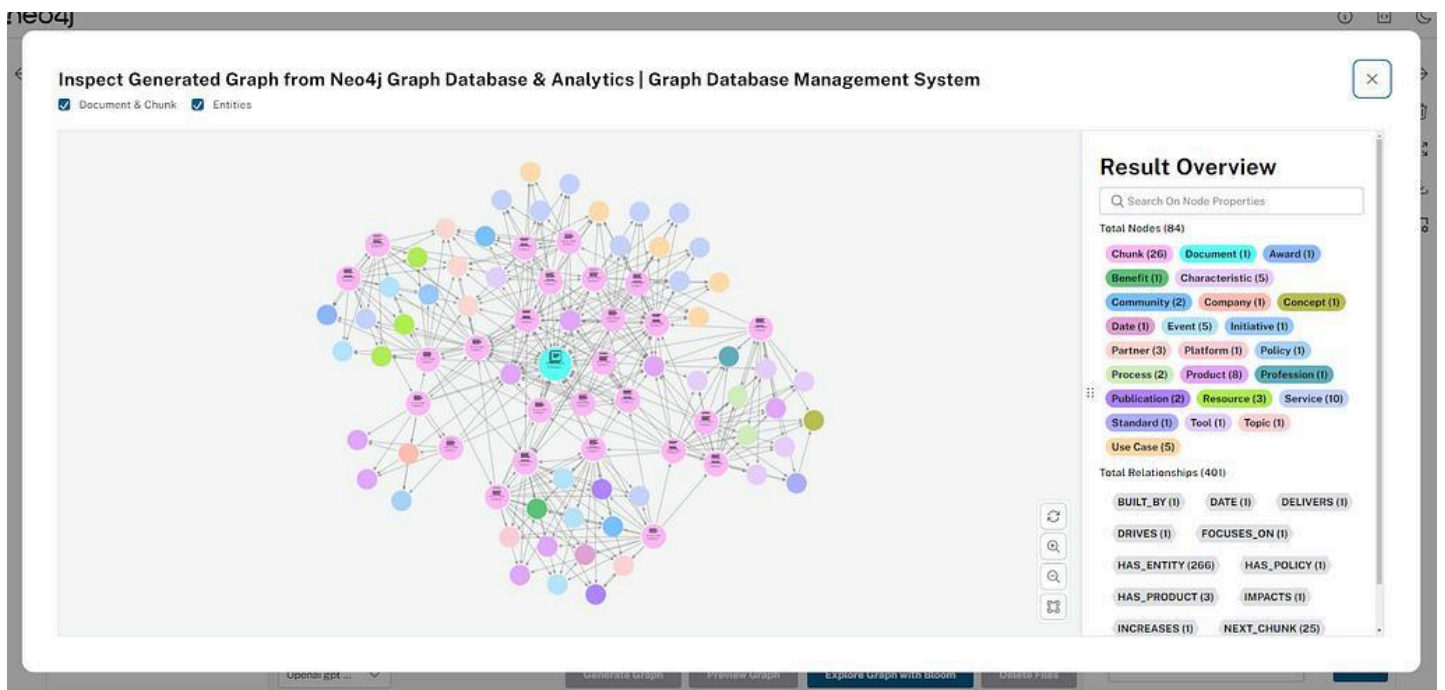
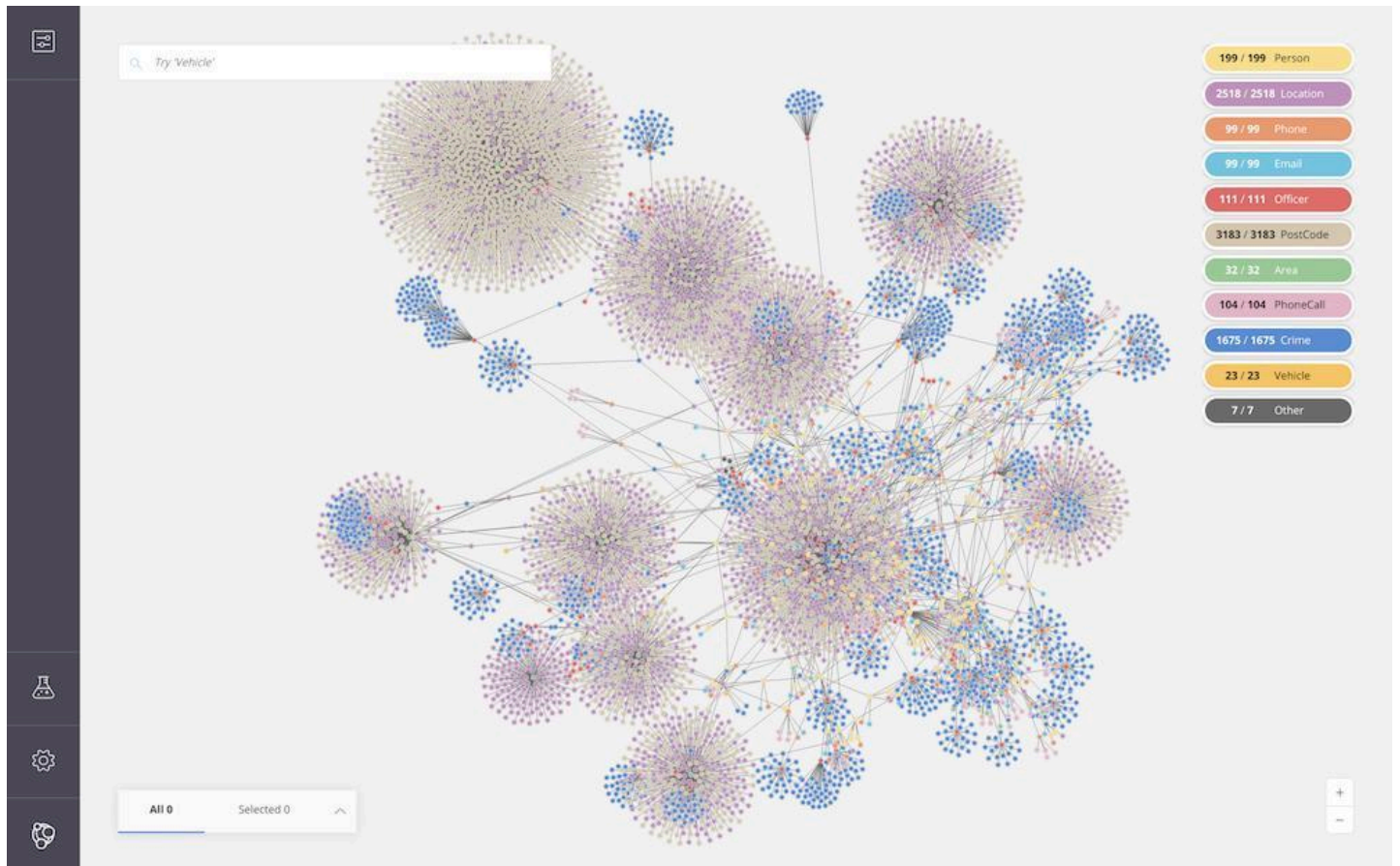


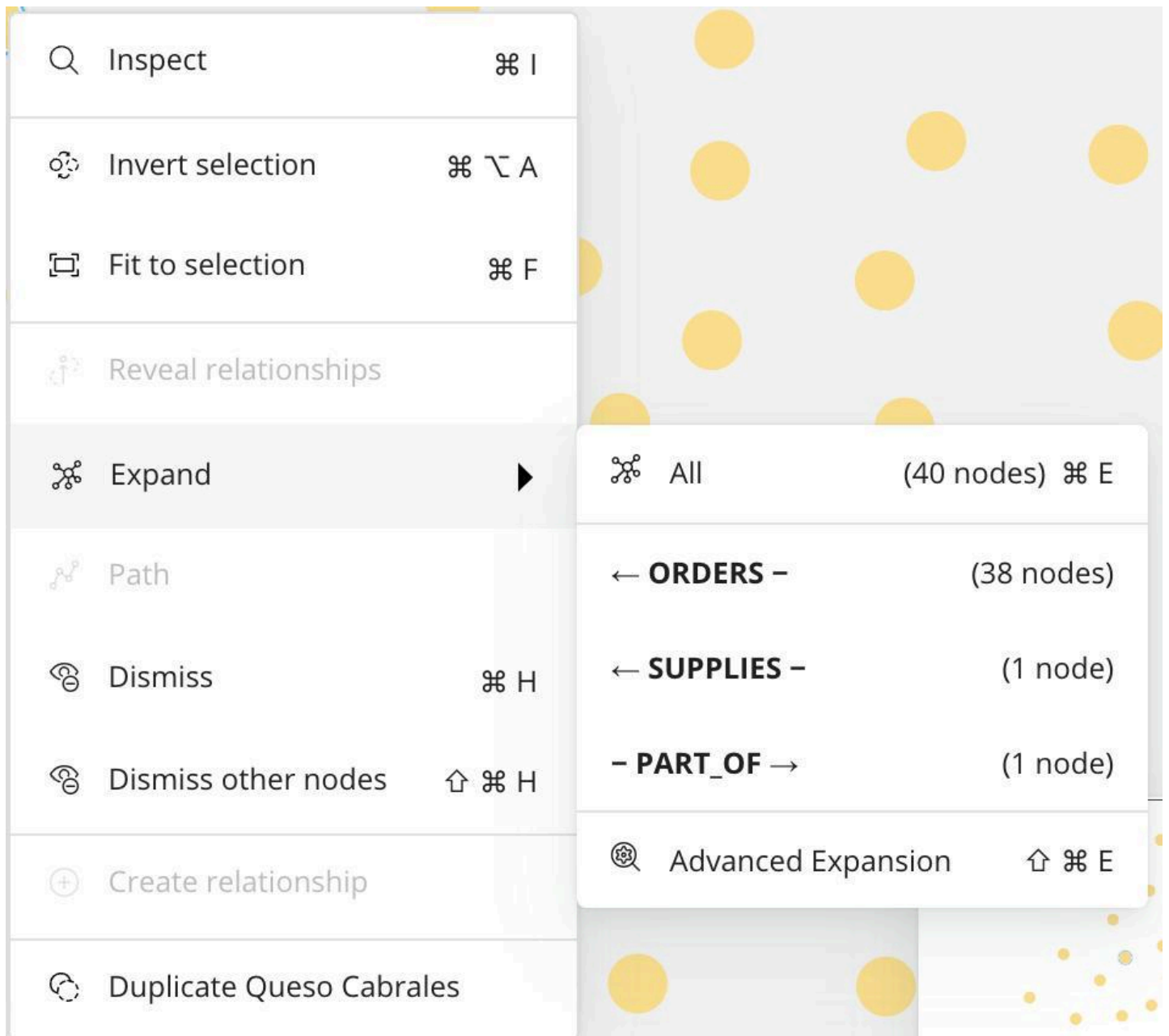


Плюси	Мінуси	Сфера
• ForceAtlas2 — один із найкращих auto-layout • Community detection • Filters • Millions of edges • Heatmaps • Clustering • Дуже хороший для “розплутування”	• Не realtime • Java desktop app • Не дуже сучасний UI	• Аналіз соцмереж • OSINT • Knowledge graph • Військові / логістичні графи

Neo4j Bloom

Офіційна visualization для Neo4j





Search



All (268) Selected (0)

Find nodes and relationships

Find nodes and relationships

All (268) Nodes (114) Relationships (154)

Supplier 29

Charlotte Cooper	country	UK
	address	49 Gilbert St.
	contactTitle	Purchasing Manager
Supplier	supplierID	1

Product 77

Chang	unitPrice	19
	reorderLevel	25
Product	unitsInStock	17
	productID	2

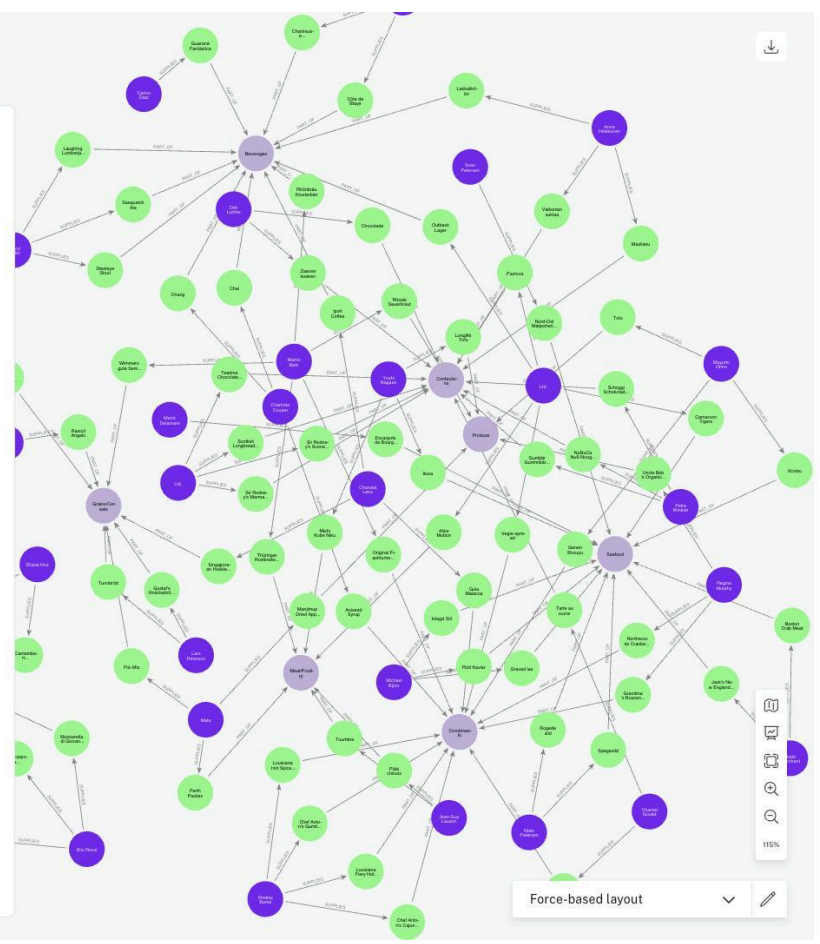
Category 8

Beverages	description	Soft drinks, coffees, teas, beers, and ales
	categoryName	Beverages
Category	categoryID	1

PART_OF 77

Chang	PART_OF	Beverages
Product		Category

SUPPLIES 77





Search graph



Product

Category

Supplier

Customer

Order

(any)

(any)

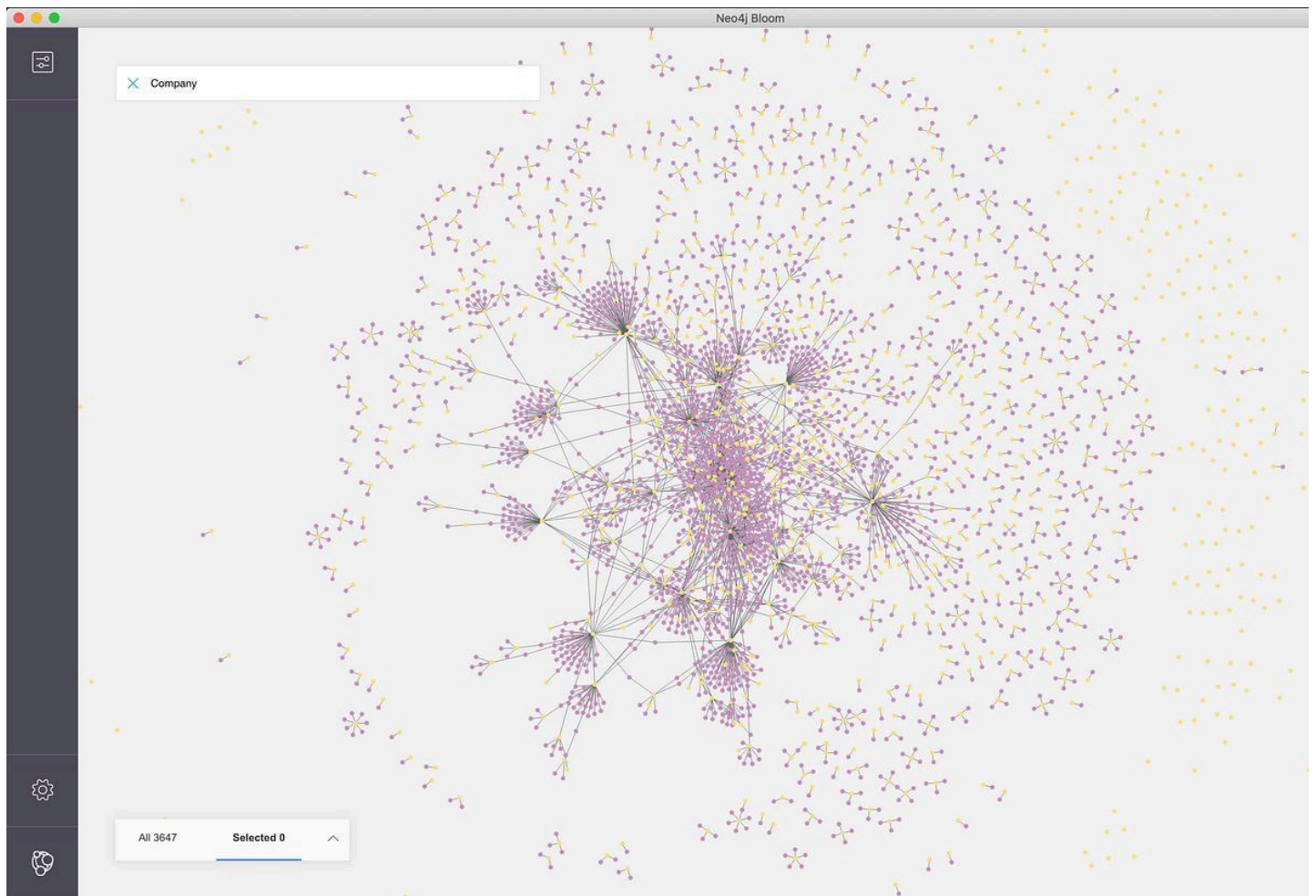
ORDERS

(any)

PART_OF

(any)

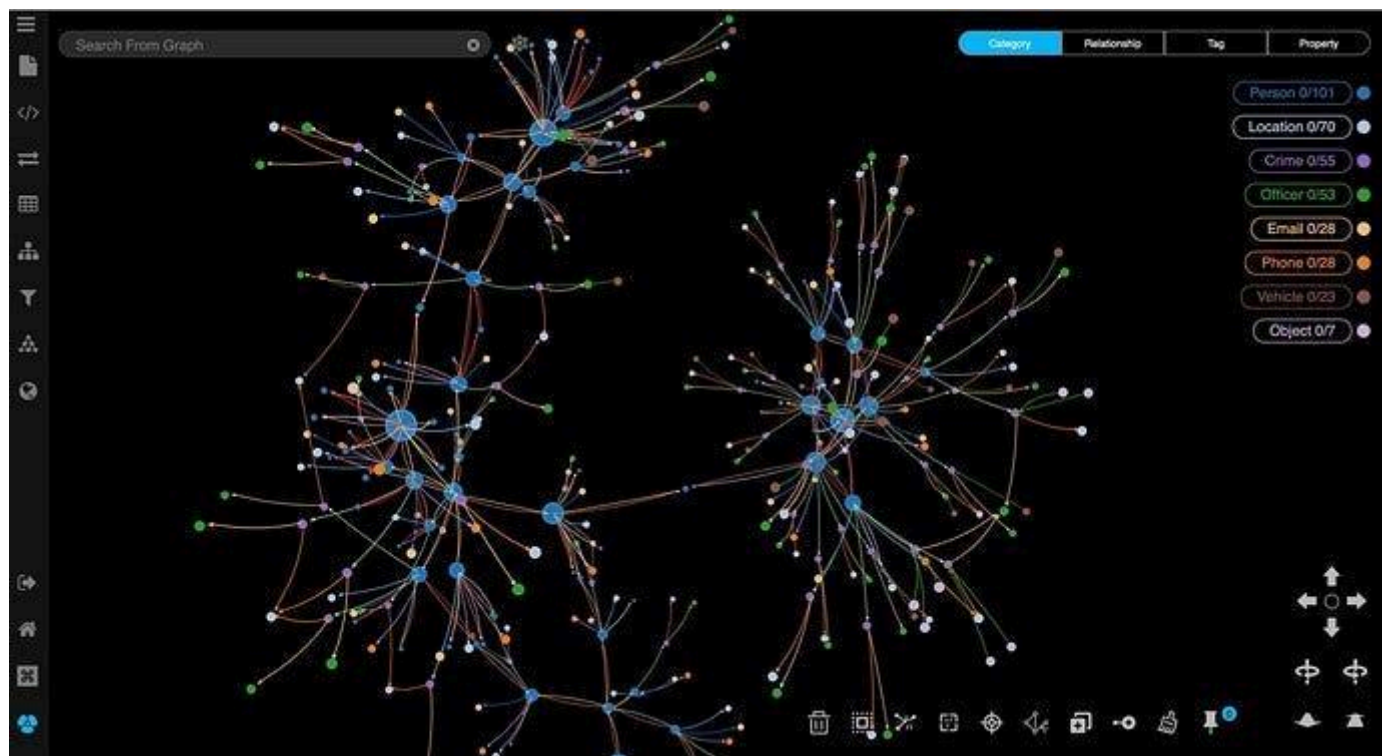
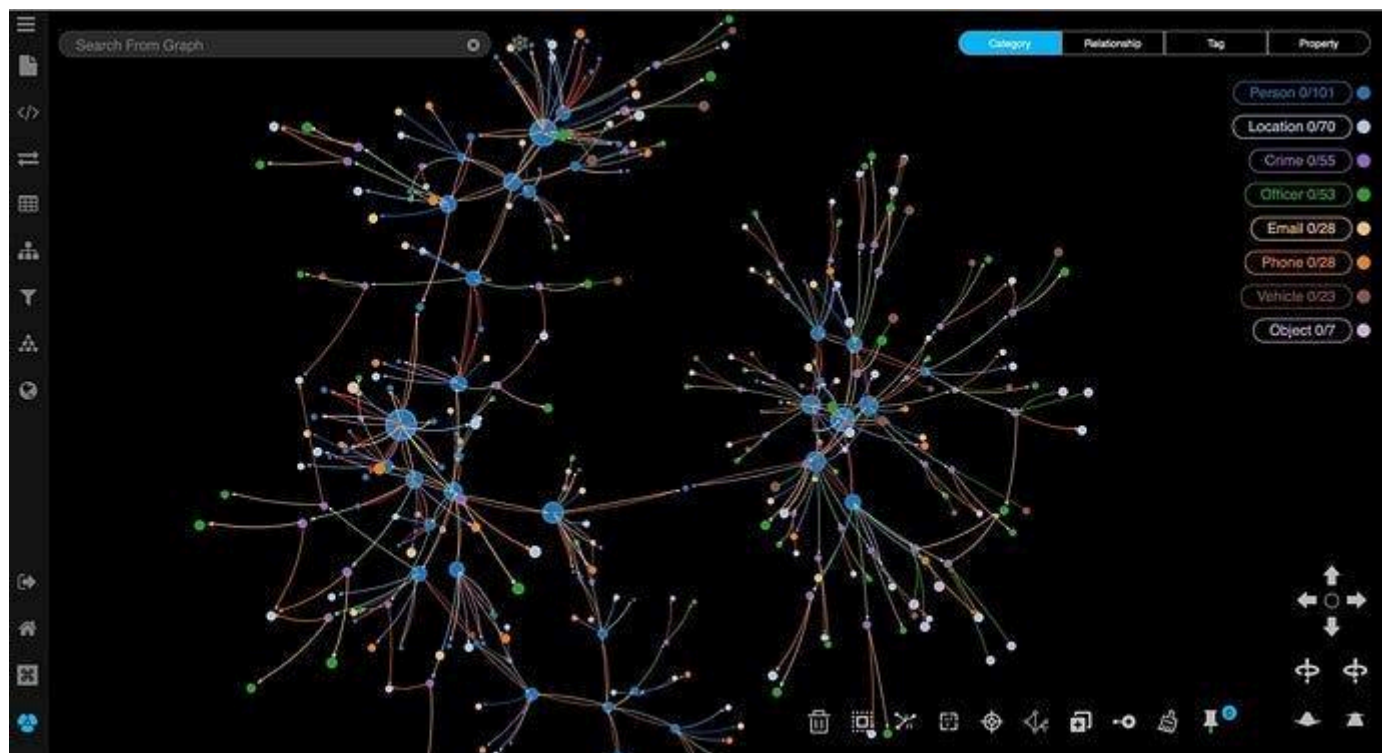
PURCHASED



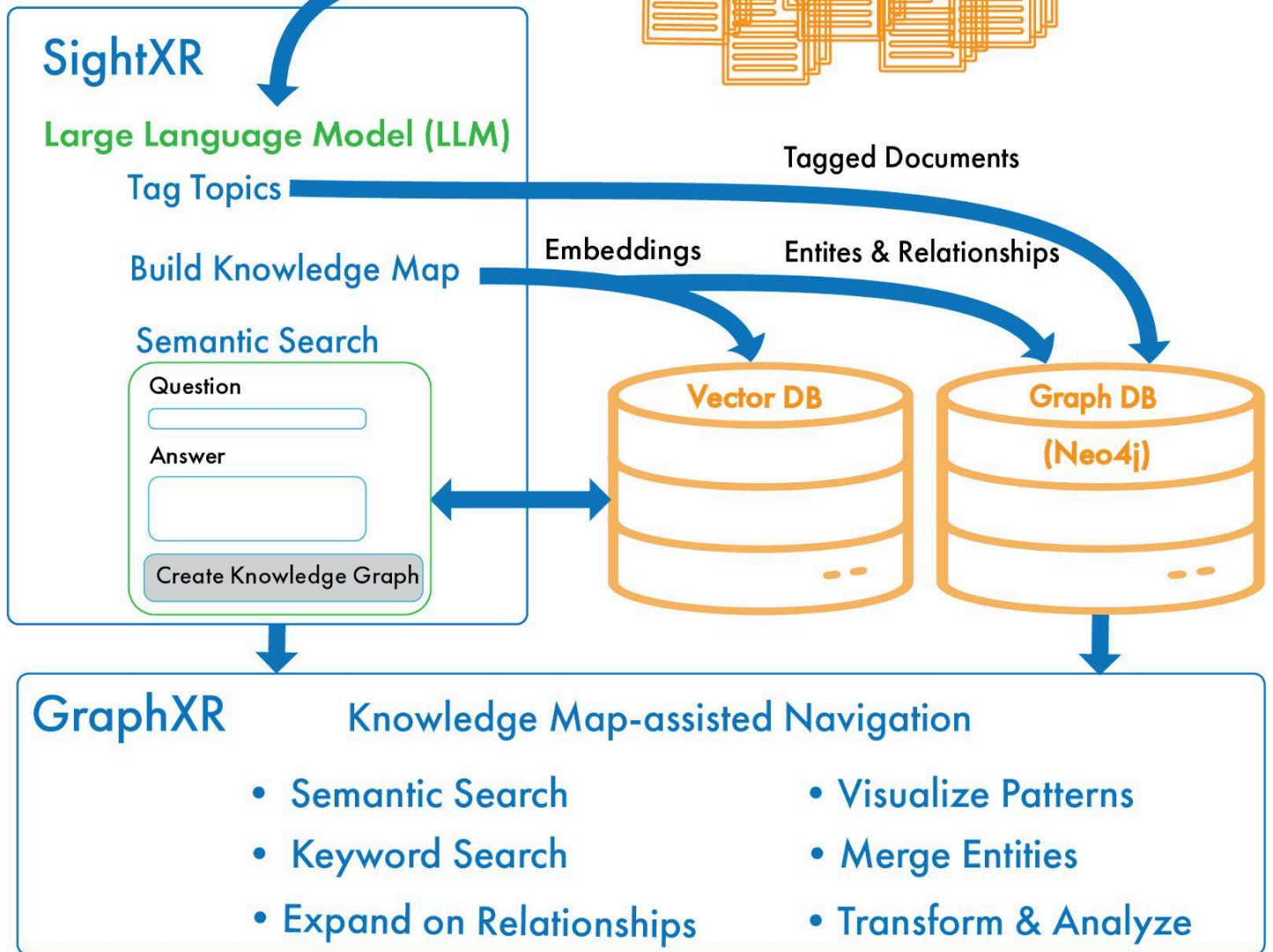
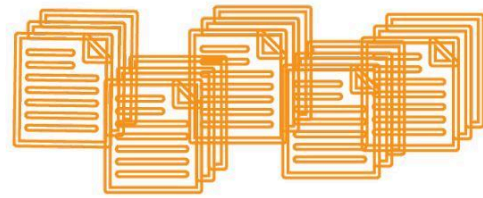
Плюси	Мінуси
• Natural language search • Красивий UI • Dynamic expansion • Styling • Дуже зручно для exploration	• Для REALLY huge graph слабуватий • Enterprise features

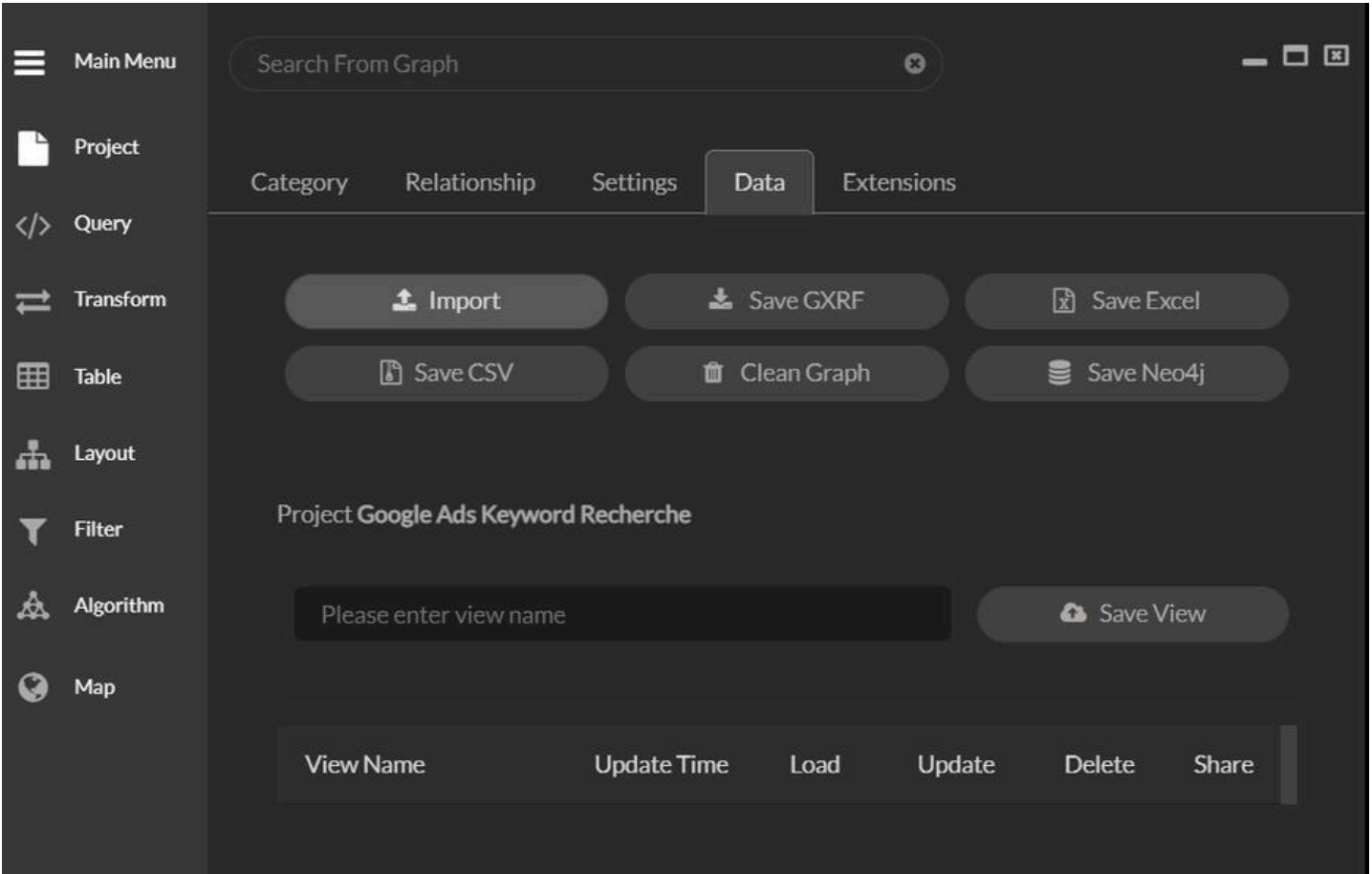
GraphXR

Дуже потужна enterprise visualization



Unstructured Documents

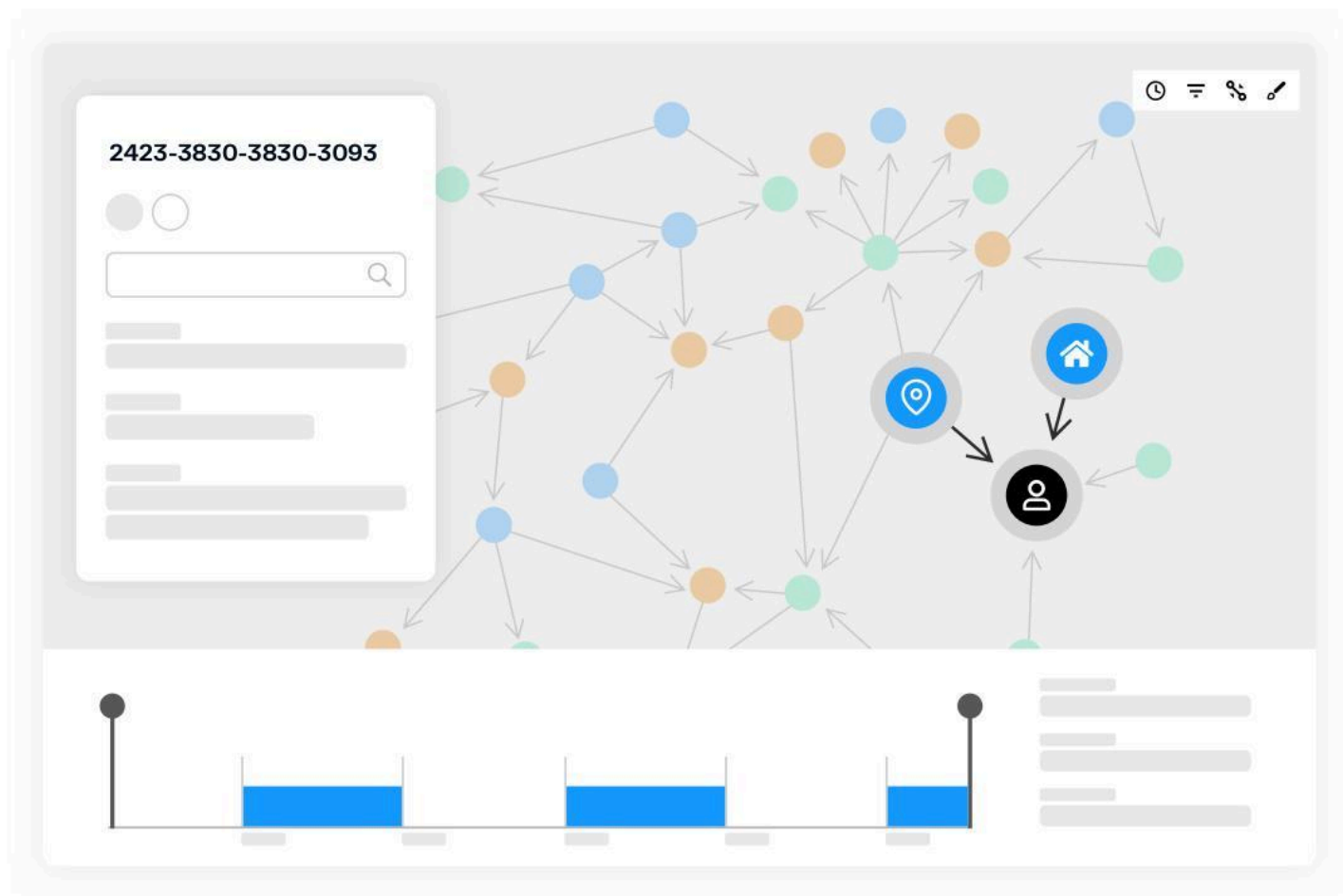




Плюси	Сфера
• Gigantic graphs • GPU rendering • Semantic exploration • AI/knowledge graphs • Dynamic filtering	• Кібербезпеки • Fraud detection • Розвідки • Enterprise graph analytics

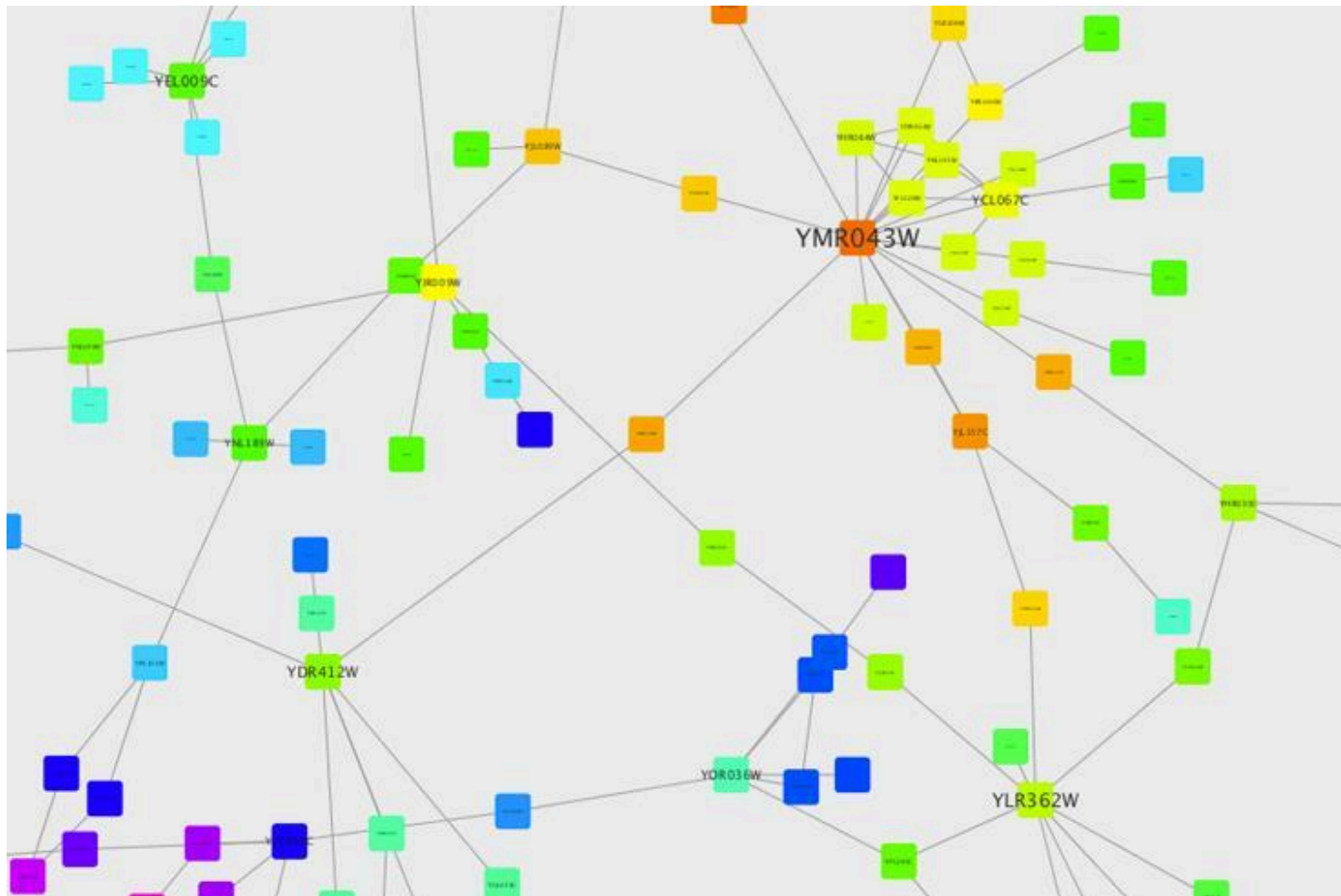
Linkurious Enterprise

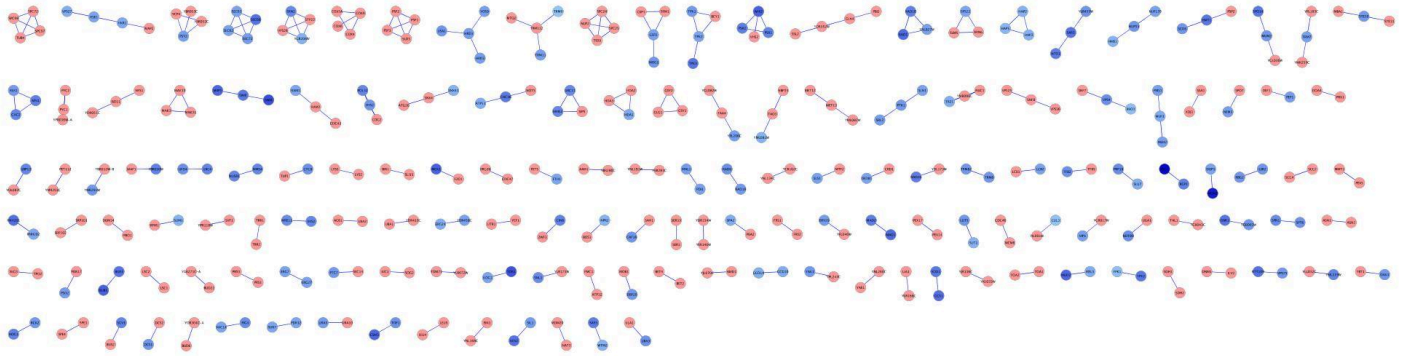
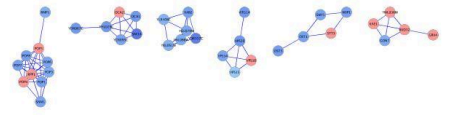
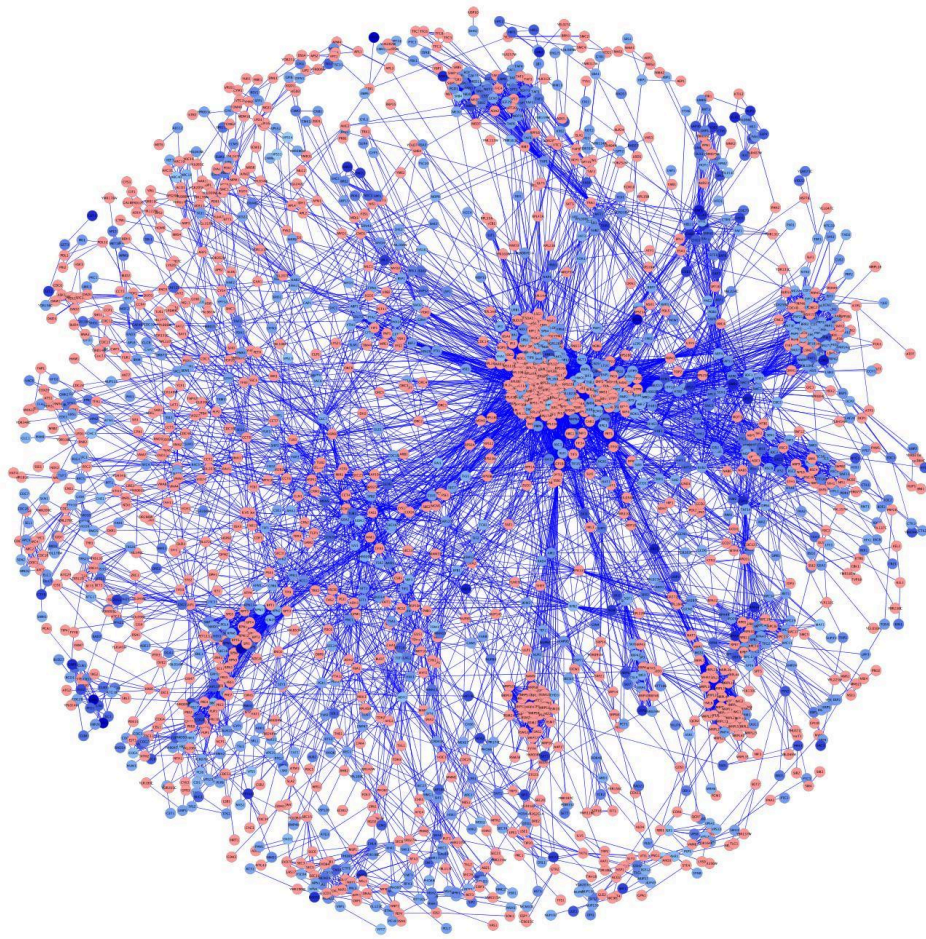
Дуже популярний для OSINT / security / investigations



Cytoscape.js

Найкраща JS-бібліотека для custom graph UI





Cytoscape.js

js.cytoscape.org

Find sections

Demos

Introduction

Factsheet

About

Packages

Releases

Citation

Funding

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Graph model

Architecture & API

Functions

Object ownership

Gestures

Position

Elements JSON

Compound nodes

Getting started

Including Cytoscape.js

Initialisation

Specifying basic options

Next steps

Core

Initialisation

Graph manipulation

cy.add()

cy.remove()

cy.collection()

cy.getElementById()

cy.\$() et al

cy.batch() et al

Cytoscape.js

Graph theory (network) library for visualisation and analysis

Repo

Github

Updates

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News & tutorials

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Questions

StackOverflow

Ask a question

StackOverflow

Community discussions

Github

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Cite

Oxford Bioinformatics Article

npm

v3.18.0

Download

v3.18.0

Extensions

60

npm installs

208k/month

master branch

passing

unstable branch

passing

Demos

Cola.js gene-gene graph

Tokyo railways

Wine & cheese

SBGN stylesheet

Popper.js extension

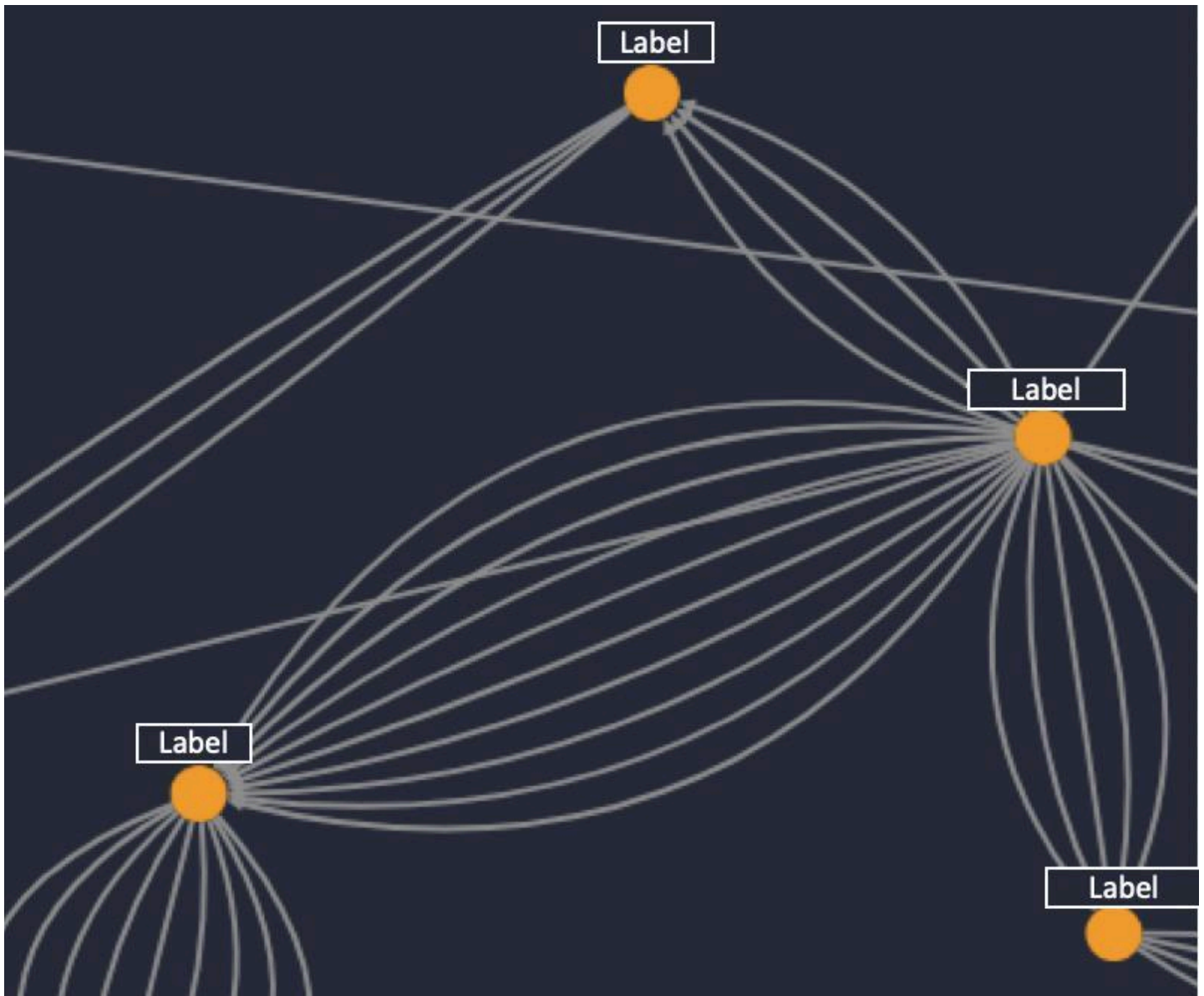
Popper.js extension &

Automove extension

Cxtmenu extension

Edgehandles extension

Compound drag-and-



cytoscape-fcose demo (constraint)

Load Graph
sample3 - relative

Randomize
fCoSE
Incremental

Draft
Transform
Enforce
CoSE

Constraints
Import
Export

Fixed Node Constraint
r1
x: 653 y: 279
Add

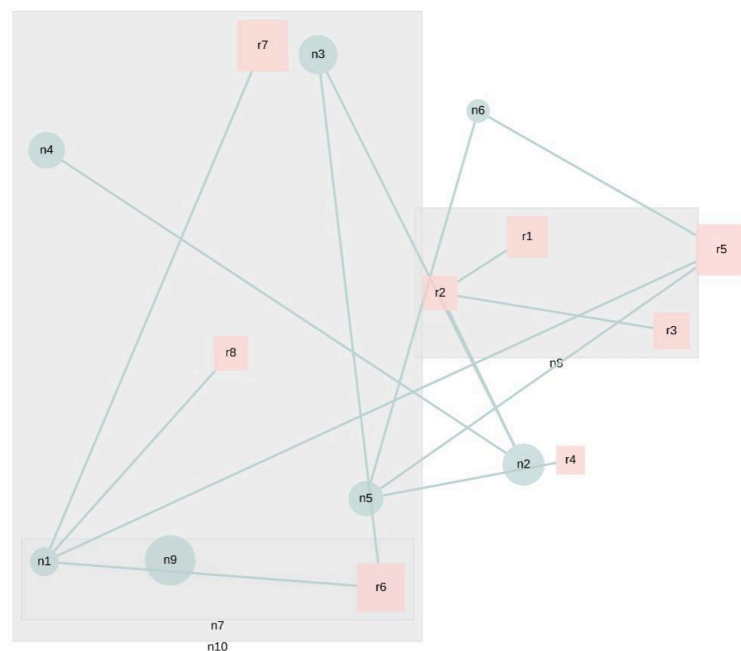
Alignment Constraint
Selected Nodes Vertically
Add
Selected Nodes Horizontally
Add
Click on a node for selecting. Shift + click for extending selection.

Relative Placement Constraint
r1
left-right
Gap:
Add
r1

Hover a constraint row to see involved nodes.

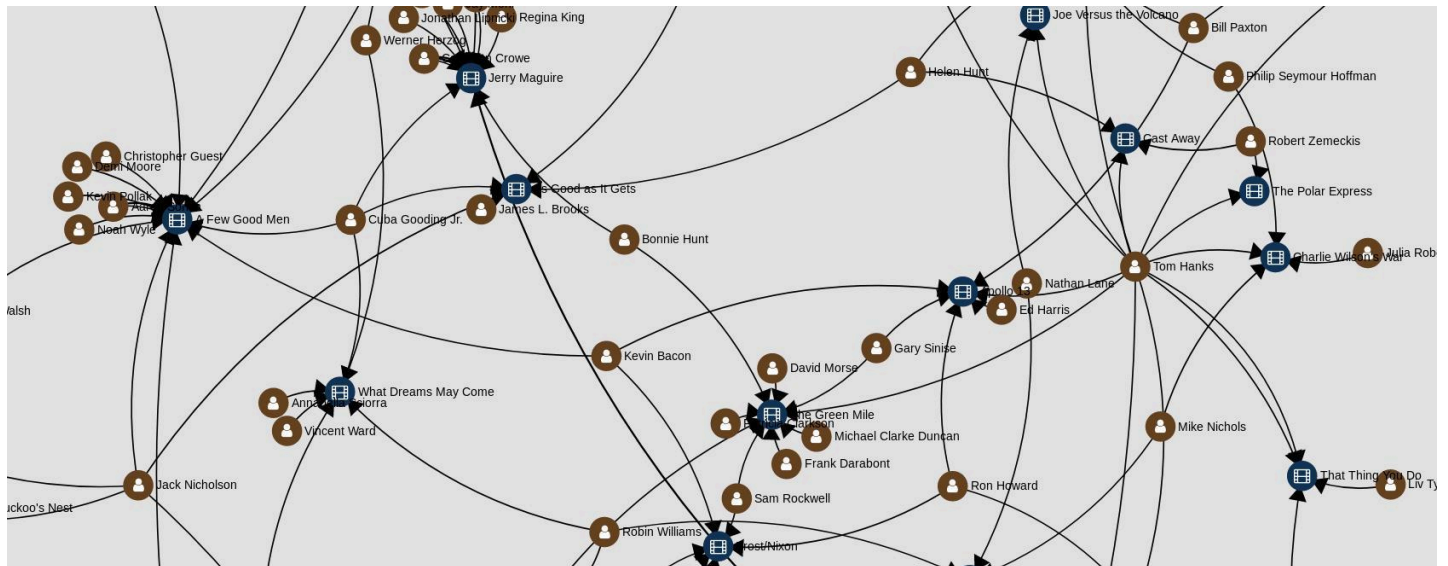
Type	Nodes	Info
Relative	r1 - r2	l-r - 100
Relative	r1 - r2	t-b - 100
Relative	r2 - r3	l-r - 200
Relative	r4 - r5	t-b - 150
Relative	r6 - r7	l-r - 150
Relative	r8 - r7	t-b - 100

4



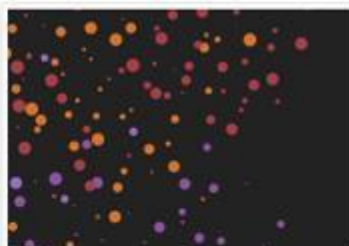
Sigma.js

Ultra-fast WebGL graph rendering



Load and display a GEXF encoded graph

An example showing how to use the `sigma.parseGexf.js` plugin.



ForceAtlas2 on a random graph

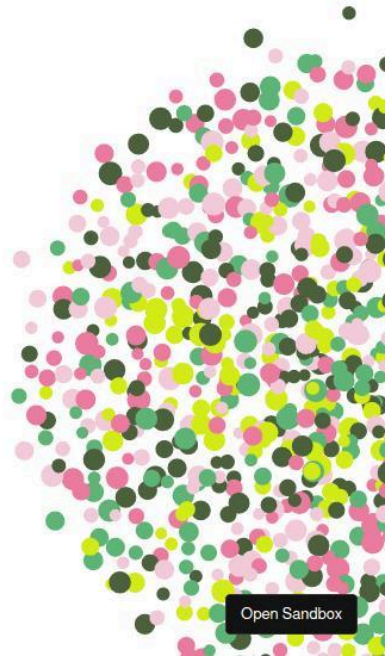
An example showing how to use the `sigma.forceatlas2.js` plugin, on a random graph.



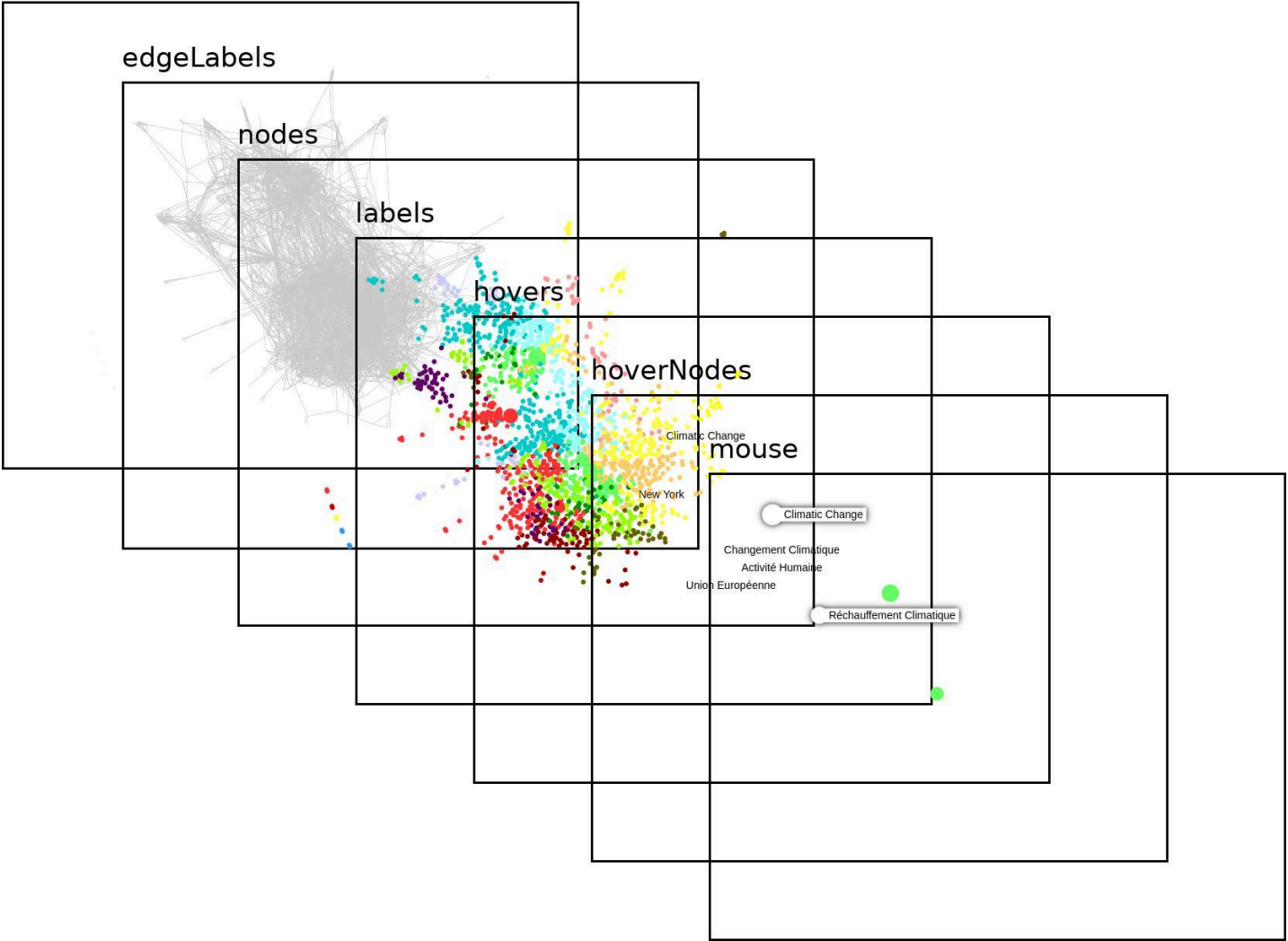
Parse graph data / use custom events

An example showing how to parse the graph data dynamically.

Sigma Js



edges





TUTORIAL

v1.2.0

DOWNLOAD



Sigma is a JavaScript library **dedicated to graph drawing**. It makes easy to publish networks on Web pages, and allows developers to integrate network exploration in rich Web applications.

Get started with sigma

The following example shows how simple it is to use sigma to display a JSON encoded graph file.

DATA

HTML

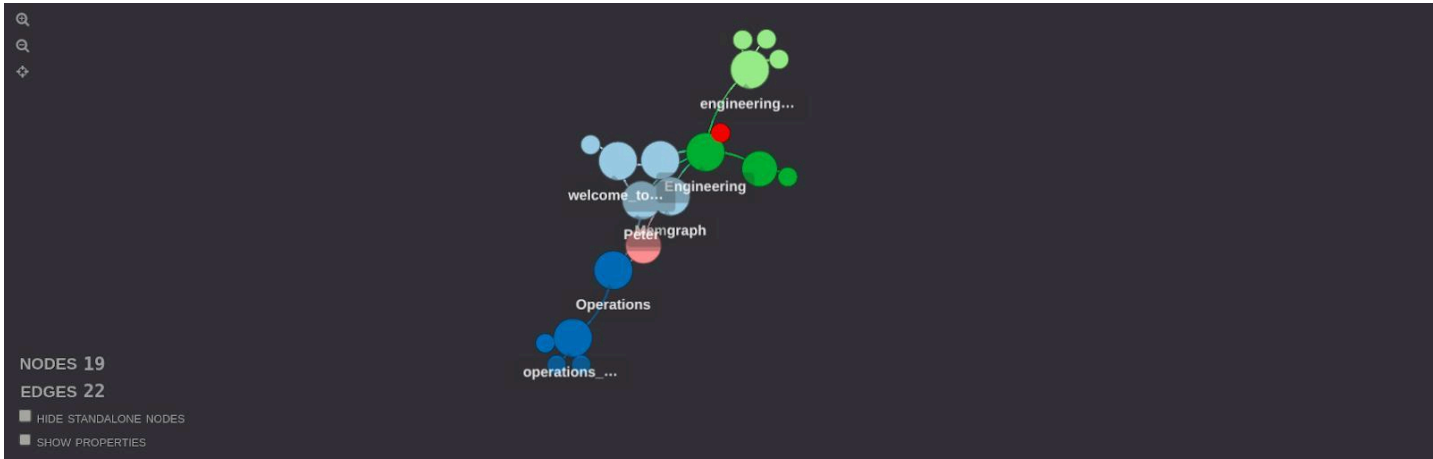
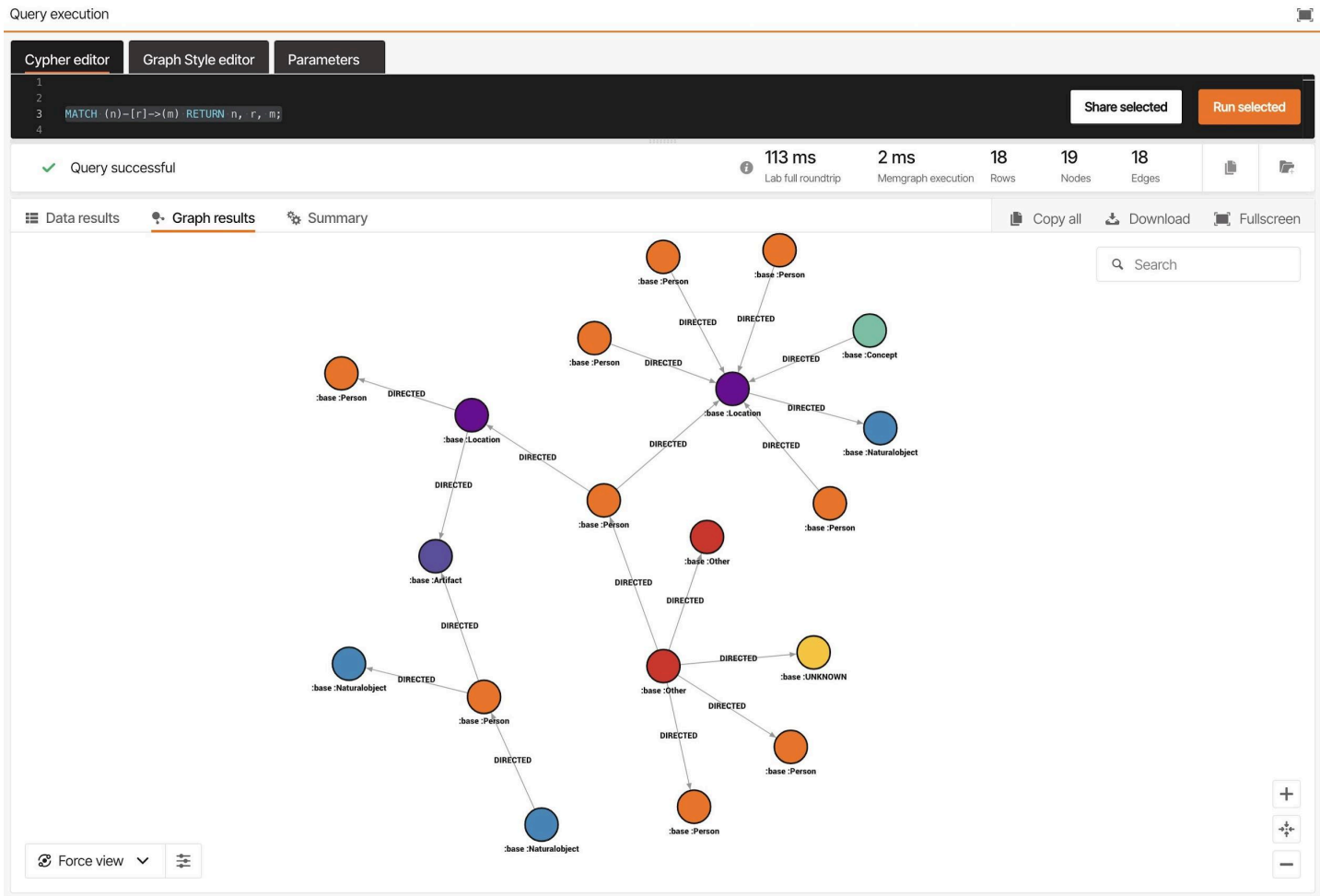
RESULT

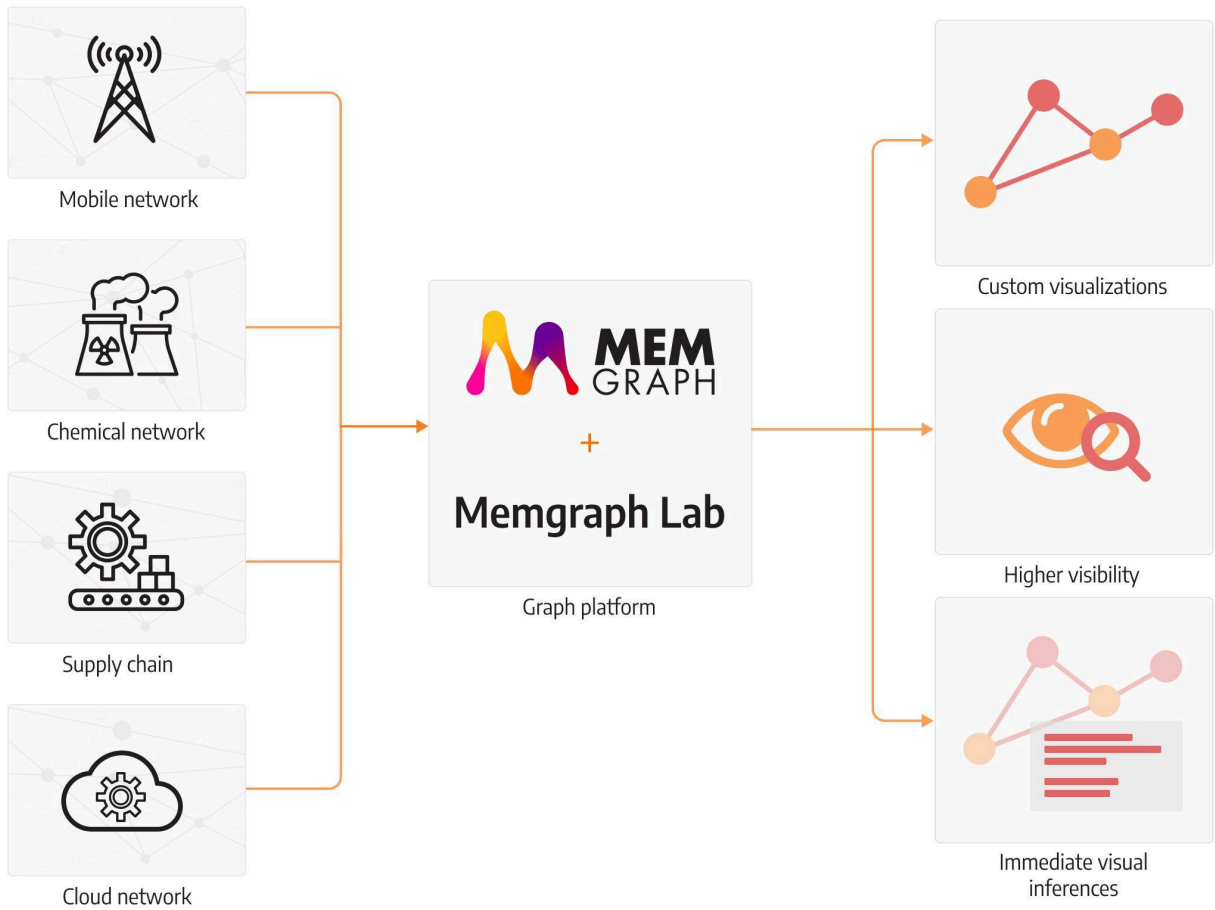
web programming group
happyco

Плюси	Мінуси
- Дуже швидкий - Great for huge graphs - WebGL	Менше built-in analytics

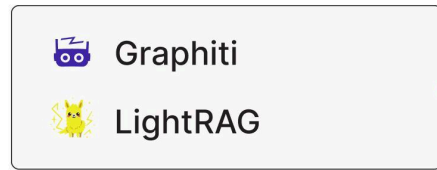
Memgraph Lab

Neo4j-like visualization + realtime graph + AI/Knowledge Graph напрямую

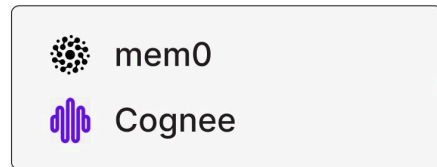




Knowledge Graph Creation



Agentic Memory



Agentic Frameworks

